

Julia DiTomas

juliaditomas@gmail.com

https://juliaditomas.github.io/personal_site/

Interests: Construction, automation, radio, optics, remote sensing, communication protocols, error correction, space, mathematics
Skills: Python, SQL, C/C++, Matlab, Linux, UI design, Arduino, Altium Designer, AutoCAD, MicroStation, data analysis, RF analysis, AWR Microwave Office, analog circuits, debugging/troubleshooting, technical writing, SIMetrix, FEKO, Verilog

Education

University of Colorado Boulder, Bachelor of Science

Major: Electrical and Computer Engineering

Minor: Computer Science

Work Experience

Kiewit Infrastructure Engineers: Jr. Communication Systems Engineer -- May 2025 - present

- Produced security, network, and SCADA system drawings for projects under contract to meet client requirements.
- Calculated loads and identified supplementary components required to support communications equipment.
- Coordinated with other disciplines, stakeholders, and vendors to ensure functionality, constructibility, and optimal designs.
- Supported Kiewit Power Engineering estimating by producing preliminary designs and equipment counts for power plants.
- Participated in the quality control process to validate designs. Reviewed designs by other team members and feedback left on my drawings. Updated plans as necessary to submit to clients on time.
- Ran wires and installed new components in the Comms Lab.

Maxar Technologies: RSS Absolute Radiometric Calibration Technician -- May 2023 - May 2025

- Collected field data to perform and improve each sensor's calibration.
- Wrote scripts for automatic calculations of satellite viewing angles and calibration coefficients.
- Maintained field equipment such as solar panels, wind turbine, sun photometer, and weather station.
- With fellow interns, designed and built a custom camera system to monitor weather at our field site and automatically upload images to AWS during calibration data acquisition.
- Developed API endpoints to streamline secure access to the calibration team's database.
- Built an interactive Python Dash app for the radiometric calibration team to visualize image orders and deliveries.
- Built a Python Dash app to monitor and display the stability and radiometric health of each sensor, and set up automatic report generation using Linux cron.

Julia's Painted Pet Art: Artist and Owner -- May 2021 - August 2024

- Designed and built a website to support a custom pet portrait business.
- Completed commissions in oils, acrylics, watercolors, and pencils as requested by customers.

University of Colorado: Discovery Learning Apprentice -- August 2022 - May 2023

- Conducted research for LunaNet project, simulating an emergency alert system for astronauts on or near the Moon and comparing transmission with different modulation schemes.
- Designed packets, built GNU Radio flowgraphs, and wrote processing scripts in Python.
- Collected data and reported findings and recommendations to the team.

University of Colorado: Teaching Facilitator -- August 2022 - December 2022

- Hosted office hours, answered questions, and assisted students in ASEN 1320, a C++ and Matlab introduction course for aerospace engineers.

Projects

Environmental Variance Analyzer (EVA) -- September 2024 - April 2025

- For our senior capstone project, my team and I developed a battery-powered environmental sensing module to support NASA's STELLA project. The EVA pod runs autonomously in the outdoor region of interest, collecting data such as air temperature, pressure, and soil moisture, and allows the user to upload the data to the EVA website for later viewing and sharing publicly.

Digital Tilt Maze Game -- March 2024 - May 2024

- In Real-Time Operating Systems (ECEN 3753) I programmed an STM32F4 microcontroller to play a game in which the user tilts the board to move a ball through a maze.

Remote-Controlled Robot with Custom Controller -- August 2023 - December 2023

- In Electronics Design Lab (ECEN 2270), I built a remote-controlled robot that drives around and plays a selection of melodies, utilizing Arduino, Bluetooth Low Energy, and feedback control loop circuitry.

Repurposed Materials Antenna Build and Analysis -- April 2024 - May 2024

- In Electromagnetic Waves and Transmissions (ECEN 3410), I teamed up with a partner and built an antenna out of semi-random material (a wine rack) bought from the thrift store, modeled and simulated it using FEKO software, and compared our results with vector network analyzer measurements.